Salima Lamsiyah

List of Publications

Book in Preparation

B1. **Salima Lamsiyah**. Responsible Natural Language Processing: Meaning, Power, and Accountability in Language Technologies. *John Benjamins Publishing Company*. **Book in preparation under contract; expected publication July 2027.**

Journal Papers

J1. Azzedine Aftiss, **Salima Lamsiyah**, Christoph Schommer, and Said Ouatik El Alaoui. 2026, HyBioSum: A Hybrid Framework for Biomedical Long-Document Summarization with Supervised Extractive and LLM-based Generation. *IEEE Access*.

J2. Aria Nourbakhsh, **Salima Lamsiyah**, Adelaide Danilov, and Christoph Schommer. 2026. Evaluating Explainable AI Attribution Methods in Neural Machine Translation via Attention-Guided Knowledge Distillation. *Natural Language Processing*, 32(3), Cambridge University Press. `https://doi.org/10.1017/nlp.2026.10028`

J3. **Salima Lamsiyah**. 2026. The Same Model, Different Responsibilities: Responsible NLP under the EU AI Act. *Natural Language Processing*, First View, pp. 1–14, Cambridge University Press. `https://doi.org/10.1017/nlp.2026.10036`

J4. **Salima Lamsiyah**. 2026. Responsible NLP: Why Performance Is No Longer Enough. *Natural Language Processing*, 32(4):518–524, Cambridge University Press. `https://doi.org/10.1017/nlp.2026.10030`

J5. Azzedine Aftiss, **Salima Lamsiyah**, Christoph Schommer, and Said Ouatik El Alaoui. 2026. A Hybrid Method for Biomedical Long-Document Summarization Using an Ensemble Extraction Approach and a Transformer-Based Model. *International Journal of Data Science and Analytics*, 22, article 140. `https://doi.org/10.1007/s41060-026-01119-4`

J6. Abdelkader El Mahdaouy, **Salima Lamsiyah**, Meryem Janati Idrissi, Hamza Alami, Zakaria Yartaoui, and Ismail Berrada. 2026. DomURLs_BERT: Pre-trained BERT-based Model for Malicious Domains and URLs Detection and Classification. *Journal of Network and Systems Management*, 34, article 36. `https://doi.org/10.1007/s10922-025-10010-9`

J7. Keerthana Murugaraj, **Salima Lamsiyah**, Marten During, and Martin Theobald. Automating Historical Insight Extraction from Large-Scale Newspaper Archives via Neural Topic Modeling. *Digital Scholarship in the Humanities*, Oxford University Press. **Forthcoming**.

J8. Keerthana Murugaraj, **Salima Lamsiyah**, Marten During, and Martin Theobald. 2025. Topic-RAG for Historical Newspapers: Enhancing Information Retrieval in Humanities Research through Topic-Based Retrieval-Augmented Generation. *Computational Humanities Research*, 1, e15. `https://doi.org/10.1017/chr.2025.10018`

J9. Keerthana Murugaraj, **Salima Lamsiyah**, and Christoph Schommer. 2025. Abstractive Summariza-

tion of Historical Documents: A New Dataset and Novel Method Using a Domain-Specific Pretrained Model. *IEEE Access*. https://ieeexplore.ieee.org/abstract/document/10838535

J10. Samir El-amrany, **Salima Lamsiyah**, Matthias R. Brust, and Pascal Bouvry. 2025. GuardHarMem and HarMDetect: A Multimodal Dataset and Benchmark Model for Fine-Grained Harmful Meme Classification. *Social Network Analysis and Mining*, 15(1), article 63. https://doi.org/10.1007/s13278-025-01475-2

J11. Azzedine Aftiss, **Salima Lamsiyah**, Said Ouatik El Alaoui, and Christoph Schommer. 2024. BioMDSum: An Effective Hybrid Biomedical Multi-Document Summarization Method Based on PageRank and Longformer Encoder-Decoder. *IEEE Access*, 12:188013–188031. https://ieeexplore.ieee.org/abstract/document/10788713

J12. **Salima Lamsiyah**, Abdelkader El Mahdaouy, and Christoph Schommer. 2023. Can Anaphora Resolution Improve Extractive Query-Focused Multi-Document Summarization? *IEEE Access*, 11:99961–99976. https://ieeexplore.ieee.org/abstract/document/10247528

J13. **Salima Lamsiyah**, Abdelkader El Mahdaouy, Saïd El Alaoui Ouatik, and Bernard Espinasse. 2023. Unsupervised Extractive Multi-Document Summarization Method Based on Transfer Learning from BERT Multi-Task Fine-Tuning. *Journal of Information Science*, 49(1):164–182. https://doi.org/10.1177/0165551521990616

J14. **Salima Lamsiyah**, Abdelkader El Mahdaouy, Said Ouatik El Alaoui, and Bernard Espinasse. 2023. Unsupervised Query-Focused Multi-Document Summarization Based on Transfer Learning from Sentence Embedding Models, BM25 Model, and Maximal Marginal Relevance Criterion. *Journal of Ambient Intelligence and Humanized Computing*, 14(3):1401–1418. https://doi.org/10.1007/s12652-021-03165-1

J15. **Salima Lamsiyah**, Abdelkader El Mahdaouy, Bernard Espinasse, and Saïd El Alaoui Ouatik. 2021. An Unsupervised Method for Extractive Multi-Document Summarization Based on Centroid Approach and Sentence Embeddings. *Expert Systems with Applications*, 167, article 114152. https://www.sciencedirect.com/science/article/pii/S0957417420308952

Conference and Workshop Papers

C1. Oleksandr Marchenko Breneur, Adelaide Danilov, Aria Nourbakhsh, and **Salima Lamsiyah**. 2026. NOTAI.AI: Explainable Detection of Machine-Generated Text via Curvature and Feature Attribution. *EMNLP 2026*. **Accepted; forthcoming.** https://arxiv.org/abs/2603.05617

C2. **Salima Lamsiyah**. 2026. FairDetect: Training-Time Fairness for AI-Generated Text Detection via Perplexity-Adaptive Focal Loss. *Artificial Intelligence in Education (AIED 2026), Part III*, pp. 398–406. Springer. https://doi.org/10.1007/978-3-032-29760-0_44

C3. **Salima Lamsiyah**. 2026. SPARK: A Student-Centered Framework for Teaching Natural Language Processing in the Age of Large Language Models. *Artificial Intelligence in Education (AIED 2026), Part V*, pp. 554–563. Springer. https://doi.org/10.1007/978-3-032-29770-9_60

C4. Keerthana Murugaraj, **Salima Lamsiyah**, and Martin Theobald. 2026. RAGVUE: A Diagnostic View for Explainable and Automated Evaluation of Retrieval-Augmented Generation. *Proceedings of EACL 2026, Volume 3*, pp. 512–526. Association for Computational Linguistics.

C5. Ernesto L. Estevanell-Valladares, **Salima Lamsiyah**, Alicia Picazo-Izquierdo, Tharindu Ranasinghe, Ruslan Mitkov, and Rafael Muñoz. 2026. SmartMatch: Real-Time Semantic Retrieval for Trans-

lation Memory Systems. *Proceedings of EACL 2026, Volume 3*, pp. 527–536. Association for Computational Linguistics.

C6. Oleksandr Marchenko, Adelaide Danilov, Aria Nourbakhsh, and **Salima Lamsiyah**. 2026. CSULoRA: Closest Safe Update Low-Rank Adaptation. *Proceedings of the Second International Conference on Natural Language Processing and Artificial Intelligence for Cyber Security (NLPAICS 2026)*, pp. 103–112. University of Alicante. https://aclanthology.org/2026.nlpaics-1.11/

C7. Christophe Friezas Gonçalves, **Salima Lamsiyah**, and Christoph Schommer. 2026. LuxDiagRC: A Diagnostic Reading Comprehension Corpus for Luxembourgish with Linguistic and Cognitive Annotation Layers. *Proceedings of the Second Workshop on Language Models for Low-Resource Languages (LoResLM 2026)*, EACL. Association for Computational Linguistics. https://aclanthology.org/2026.loreslm-1.46/

C8. Vlada Khomenko, Christoph Schommer, and **Salima Lamsiyah**. 2026. A Knowledge-Based Book Recommender through Enhanced Retrieval-Augmented Generation. *Proceedings of ICAART 2026, Volume 3*, pp. 2041–2049. SCITEPRESS. https://doi.org/10.5220/0014480000004052

C9. Akshata Tejas Karandikar, Christoph Schommer, and **Salima Lamsiyah**. 2026. SHAP-LLM: Explainability-Guided Synthetic Tabular Data Generation Using Large Language Models. *Proceedings of ICAART 2026, Volume 5*, pp. 4720–4727. SCITEPRESS. https://doi.org/10.5220/0014485700004052

C10. Keerthana Murugaraj, **Salima Lamsiyah**, and Martin Theobald. 2025. ExpandFuse: A Hybrid Retrieval Framework with Query Expansion and Topic-aware Reranking for Multi-Hop Question Answering. *Proceedings of the 2025 IEEE International Conference on Big Data*. IEEE.

C11. **Salima Lamsiyah**, Aria Nourbakhsh, Samir El-Amrany, and Christoph Schommer. 2025. Privacy-preserving federated learning for student dropout prediction: Enhancing model transparency with explainable AI. *International Conference on Artificial Intelligence in Education*, pp. 324–332. Springer. https://link.springer.com/chapter/10.1007/978-3-031-98465-5_41

C12. **Salima Lamsiyah**, Aria Nourbakhsh, and Christoph Schommer. 2025. Trust but verify: A comprehensive survey of faithfulness evaluation methods in abstractive text summarization. *Proceedings of the 15th International Conference on Recent Advances in Natural Language Processing (RANLP 2025)*, Varna, Bulgaria. Association for Computational Linguistics.

C13. Aria Nourbakhsh, **Salima Lamsiyah**, and Christoph Schommer. 2025. Quantifying the overlap: Attribution maps and linguistic heuristics in encoder–decoder machine translation models. *Proceedings of the 15th International Conference on Recent Advances in Natural Language Processing (RANLP 2025)*, Varna, Bulgaria. Association for Computational Linguistics.

C14. Samir El-Amrany, Matthias R. Brust, **Salima Lamsiyah**, and Pascal Bouvry. 2025. Reddit-v: A virality prediction dataset and zero-shot evaluation with large language models. *Proceedings of the 15th International Conference on Recent Advances in Natural Language Processing (RANLP 2025)*, Varna, Bulgaria. Association for Computational Linguistics.

C15. Keerthana Murugaraj, **Salima Lamsiyah**, Marten During, and Martin Theobald. 2025. Mining the past: A comparative study of classical and neural topic models on historical newspaper archives. *Proceedings of the 5th International Conference on Natural Language Processing for Digital Humanities*, pp. 452–463, Albuquerque, USA. Association for Computational Linguistics. https://aclanthology.org/2025.nlp4dh-1.39/

C16. **Salima Lamsiyah**, Kamyar Zeinalipour, Samir El-Amrany, Matthias Brust, Marco Maggini, Pascal Bouvry, and Christoph Schommer. 2025. ArabicSense: A benchmark for evaluating commonsense reasoning in Arabic with large language models. *Proceedings of the 4th Workshop on Arabic Corpus Linguistics (WACL-4)*, pp. 1–11, Abu Dhabi, UAE. Association for Computational Linguistics. (Best Paper Award) https://aclanthology.org/2025.wacl-1.1.pdf

C17. Azzedine Aftiss, **Salima Lamsiyah**, Christoph Schommer, and Said Ouatik El Alaoui. 2025. Empirical evaluation of pre-trained language models for summarizing Moroccan Darija news articles. *Proceedings of the 4th Workshop on Arabic Corpus Linguistics (WACL-4)*, pp. 77–85, Abu Dhabi, UAE. Association for Computational Linguistics. https://aclanthology.org/2025.wacl-1.9.pdf

C18. **Salima Lamsiyah**, Abdelkader El Mahdaouy, Aria Nourbakhsh, and Christoph Schommer. 2024. Fine-tuning a large language model with reinforcement learning for educational question generation. *International Conference on Artificial Intelligence in Education*, pp. 424–438. Springer. (Nominated for the Best Paper Award) https://link.springer.com/chapter/10.1007/978-3-031-64302-6_30

C19. Azzedine Aftiss, **Salima Lamsiyah**, Christoph Schommer, and Said Ouatik El Alaoui. 2024. Abstractive biomedical long document summarization through zero-shot prompting. *2024 Sixth International Conference on Intelligent Computing in Data Sciences (ICDS)*, pp. 1–6. IEEE. https://ieeexplore.ieee.org/abstract/document/10756301

C20. **Salima Lamsiyah**, Abdelkader El Mahdaouy, Hamza Alami, Ismail Berrada, and Christoph Schommer. 2023. UL & UM6P at ArAIEval shared task: Transformer-based model for persuasion techniques and disinformation detection in Arabic. *Proceedings of ArabicNLP 2023*, pp. 558–564, Singapore. Association for Computational Linguistics. https://aclanthology.org/2023.arabicnlp-1.55.pdf

C21. Abdelkader El Mahdaouy, **Salima Lamsiyah**, Hamza Alami, Christoph Schommer, and Ismail Berrada. 2023. UM6P & UL at WojoodNER shared task: Improving multi-task learning for flat and nested Arabic named entity recognition. *Proceedings of ArabicNLP 2023*, pp. 777–782, Singapore. Association for Computational Linguistics. https://aclanthology.org/2023.arabicnlp-1.87.pdf

C22. **Salima Lamsiyah**, Keerthana Murugaraj, and Christoph Schommer. 2023. Historical-domain pre-trained language model for historical extractive text summarization. *8th International Conference on Computer and Information Science and Technology (CIST 2023)*, London, United Kingdom. https://avestia.com/EECSS2023_Proceedings/files/paper/CIST/CIST_152.pdf

C23. **Salima Lamsiyah**, Abdelkader El Mahdaouy, Hamza Alami, Ismail Berrada, and Christoph Schommer. 2023. UL & UM6P at SemEval-2023 task 10: Semi-supervised multi-task learning for explainable detection of online sexism. *Proceedings of the 17th International Workshop on Semantic Evaluation (SemEval-2023)*, pp. 644–650, Toronto, Canada. Association for Computational Linguistics. https://aclanthology.org/2023.semeval-1.88.pdf

C24. Abdelkader El Mahdaouy, Hamza Alami, **Salima Lamsiyah**, and Ismail Berrada. 2023. UM6P at SemEval-2023 task 12: Out-of-distribution generalization method for African languages sentiment analysis. *Proceedings of the 17th International Workshop on Semantic Evaluation (SemEval-2023)*, pp. 1004–1010, Toronto, Canada. Association for Computational Linguistics. https://aclanthology.org/2023.semeval-1.138.pdf

C25. **Salima Lamsiyah** and Christoph Schommer. 2023. A comparative study of sentence embeddings for unsupervised extractive multi-document summarization. *Artificial Intelligence and Machine Learning*, pp. 78–95. Springer Nature Switzerland. (Nominated for the Best Paper Award) https://link.springer.com/chapter/10.1007/978-3-031-39144-6_6

C26. **Salima Lamsiyah**, Abdelkader El Mahdaouy, Bernard Espinasse, and Said Ouatik El Alaoui. 2020. Unsupervised query-focused multi-document summarization using usif sentence embedding model and maximal marginal relevance criterion. *International Conference on Advanced Intelligent Systems for Sustainable Development*, pp. 798–808. Springer. https://link.springer.com/chapter/10.1007/978-3-030-90639-9_66

C27. **Salima Lamsiyah**, Abdelkader El Mahdaouy, Said Ouatik El Alaoui, and Bernard Espinasse. 2019. A supervised method for extractive single document summarization based on sentence embeddings and neural networks. *International Conference on Advanced Intelligent Systems for Sustainable Development*, pp. 75–88. Springer. https://link.springer.com/chapter/10.1007/978-3-030-36674-2_8

C28. **Salima Lamsiyah**, Said Ouatik El Alaoui, and Bernard Espinasse. 2018. Résumé automatique guidé de textes: État de l'art et perspectives (guided summarization: State-of-the-art and perspectives). *Actes de la Conférence TALN, Volume 2*, pp. 55–72, Rennes, France. https://aclanthology.org/2018.jeptalnrecital-recital.5.pdf

C29. Xiaojing Zhao, **Salima Lamsiyah**, Emmanuele Chersoni, and Han Xu. 2026. Benchmarking Large Language Models on Mandarin Proverb Explanation and Contextual Matching. *International Conference on Language Technologies for Low-Resource Languages (LaTeLL 2026)*.

C30. **Salima Lamsiyah** and Ruslan Mitkov. 2026. Why Current XAI Is Not Enough for Arabic NLP: A Critical Survey of the Explainability Gap. *International Conference on Language Technologies for Low-Resource Languages (LaTeLL 2026)*. **Accepted; forthcoming.** https://arxiv.org/abs/2608.26144

C31. Keerthana Murugaraj, Hedi Tebourbi, Christophe Friezas Gonçalves, and **Salima Lamsiyah**. 2026. LuxDiag-RAG: Diagnostic Evaluation of Retrieval-Augmented Generation for Luxembourgish Reading Comprehension. *International Conference on Language Technologies for Low-Resource Languages (LaTeLL 2026)*. **Accepted; forthcoming.**

C32. Christophe Friezas Gonçalves, Hedi Tebourbi, Christoph Schommer, and **Salima Lamsiyah**. 2026. Assessment of Human-in-the-Loop Multi-LLMs for Low-Resource Educational Data Expansion. *International Conference on Language Technologies for Low-Resource Languages (LaTeLL 2026)*. **Accepted; forthcoming.**

C33. Christophe Friezas Gonçalves, Christoph Schommer, and **Salima Lamsiyah**. 2026. SimuLe-Lux: A Neuro-Symbolic Teaching System for Diagnosing L2 Reading Comprehension. *International Conference on Language Technologies for Low-Resource Languages (LaTeLL 2026)*. **Accepted; forthcoming.**

C34. Houdaifa Atou, Nouran Khallaf, **Salima Lamsiyah**, and Ruslan Mitkov. 2026. Towards Readability Assessment for Under-Resourced Arabic Dialects: A Study of Moroccan Darija. *International Conference on Language Technologies for Low-Resource Languages (LaTeLL 2026)*. **Accepted; forthcoming.**

Published Shared-Task Overview Papers

ST1. Nathaniel R. Robinson, Shahd Abdelmoneim, Anjali Kantharuban, Otba Alsboul, **Salima Lamsiyah**, Kelly Marchisio, and Kenton Murray. 2026. AMIYA Shared Task: Arabic Modeling In Your Accent at VarDial 2026. *Proceedings of the 13th Workshop on NLP for Similar Languages, Varieties and Dialects, pp. 1–17. Association for Computational Linguistics*. `https://aclanthology.org/2026.vardial-1.1/`

ST2. **Salima Lamsiyah**, Saad Ezzini, Abdelkader El Mahdaouy, Hamza Alami, Abdessamad Benlahbib, Samir El amrany, Salmane Chafik, and Hicham Hammouchi. 2025. M-DAIGT: A Shared Task on Multi-Domain Detection of AI-Generated Text. *Proceedings of the Shared Task on Multi-Domain Detection of AI-Generated Text, pp. 1–9. INCOMA Ltd.*. `https://aclanthology.org/2025.ranlp-mdaigt.1/`

ST3. Shadi Abudalfa, Saad Ezzini, Ahmed Abdelali, Hamza Alami, Abdessamad Benlahbib, Salmane Chafik, Mo El-Haj, Abdelkader El Mahdaouy, Mustafa Jarrar, **Salima Lamsiyah**, and Hamzah Luqman. 2025. The AraGenEval Shared Task on Arabic Authorship Style Transfer and AI-Generated Text Detection.In Proceedings of the Third Arabic Natural Language Processing Conference, Suzhou, China. Association for Computational Linguistics.

ST4. Saad Ezzini, Shadi Abudalfa, Maram Alharbi, Salmane Chafik, Hind Saleh Alatawi, Mo El-Haj, Ahmed Abdelali, Osamah Alnahari, and **Salima Lamsiyah**. 2026. DialectSentEval 2026: Arabic Dialect Sentiment Analysis and Swapping Shared Task. *ArabicNLP 2026, co-located with EMNLP 2026*. **Accepted; forthcoming.**

Submitted Papers and Manuscripts under Review

S1. Adelaide Danilov, Aria Nourbakhsh, Oleksandr Marchenko Breneur, and **Salima Lamsiyah**. "Many Are My Names": The Anatomy of the Assistant and Its Personas via Sparse Autoencoders. *Submitted to ICLR 2027*. `https://arxiv.org/abs/2608.07852`

S2. Aria Nourbakhsh, Adelaide Danilov, Christoph Schommer, and **Salima Lamsiyah**. AEyeDE: An Attention-Based Attribution Framework for AI-Generated Text Detection. *Under review at Transactions on Machine Learning Research (TMLR)*. `https://arxiv.org/abs/2606.00016`

S3. **Salima Lamsiyah**, Adelaide Danilov, Oleksandr Marchenko Breneur, and Aria Nourbakhsh. Sparse Autoencoders for Mechanistic Interpretability in NLP: An Evidence-Centered Survey. *Submitted to NAACL 2027*.

S4. ***Salima Lamsiyah***. Who Owns the Trajectory? Multi-Owner Federated Learning for Language-Agent Societies. *Submitted to AAMAS 2027*.

S5. ***Salima Lamsiyah***, *Benoit Alcaraz, Aria Nourbakhsh*. Can We Tell Why an Agent Disobeyed? Identifiability and Incentive-Compatible Routing in Normative Multi-Agent Systems. *Submitted to AAMAS 2027*.

S6. Salmane Chafik, **Salima Lamsiyah**, Saad Ezzini, Ayoub Nainia, Christophe Friezas Gonçalves, Abrar Alotaibi, Maram Alharbi, Alicia Picazo-Izquierdo, Nouran Khallaf, Xiaojing Zhao, Shadi Abudalfa, Nadia Ghezaiel Hammouda, Aadil Gani Ganie, and Hamzah Luqman. WISDOM: A Worldwide Multilingual Benchmark for Proverbs and Figurative Language. *Submitted to NAACL 2027*.

S7. Hedi Tebourbi, Christophe Friezas Gonçalves, and **Salima Lamsiyah**. LuxDiag Workbench: Interactive RAG and Annotation Diagnosis for Luxembourgish Reading Comprehension. *Submitted to EACL 2027*.

S8. Aria Nourbakhsh, Adelaide Danilov, Oleksandr Marchenko Breneur, and **Salima Lamsiyah**. Do Hallucination Features Exist? Sparse Feature Analysis of Multiple-Choice Answering Errors in LLMs. *Submitted to EACL 2027*.

S9. Keerthana Murugaraj, **Salima Lamsiyah**, and Martin Theobald. RECTIFY: An Interactive Workbench for Post-Evaluation RAG Diagnosis, Repair, and Verification. *Submitted to EACL 2027*. `https://arxiv.org/abs/2609.16764`